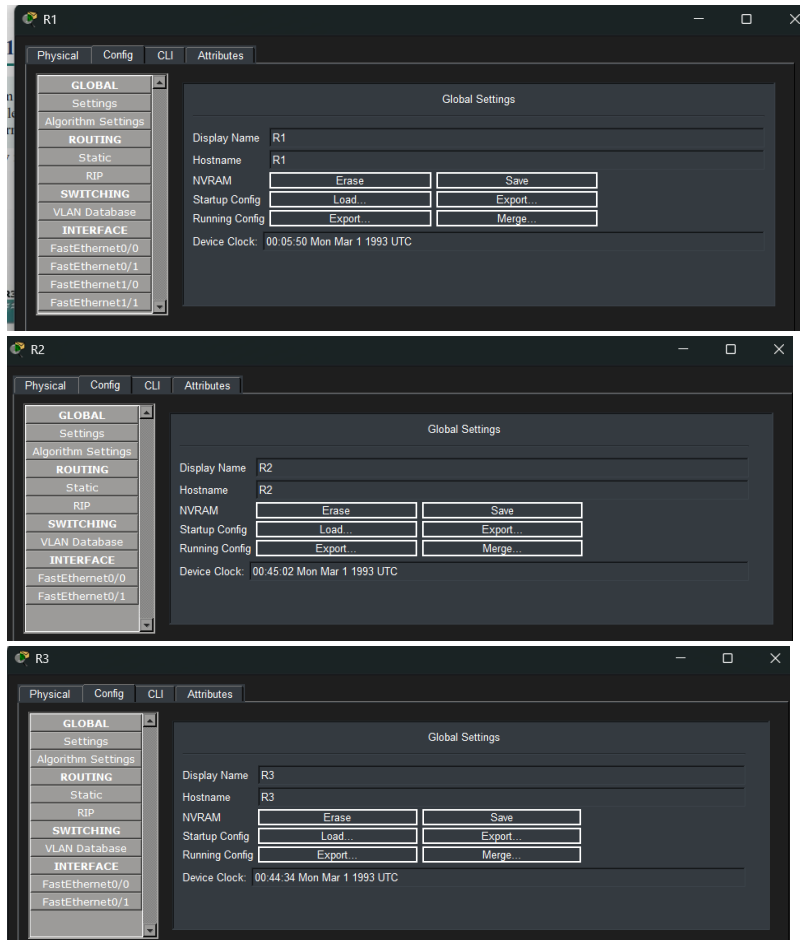
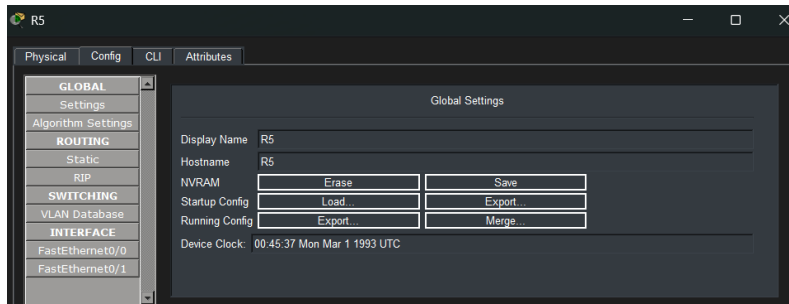
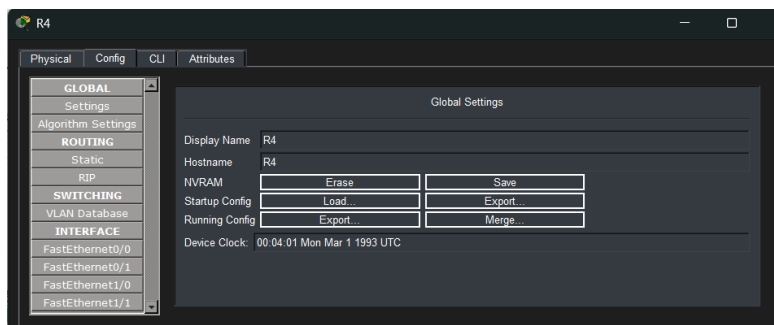


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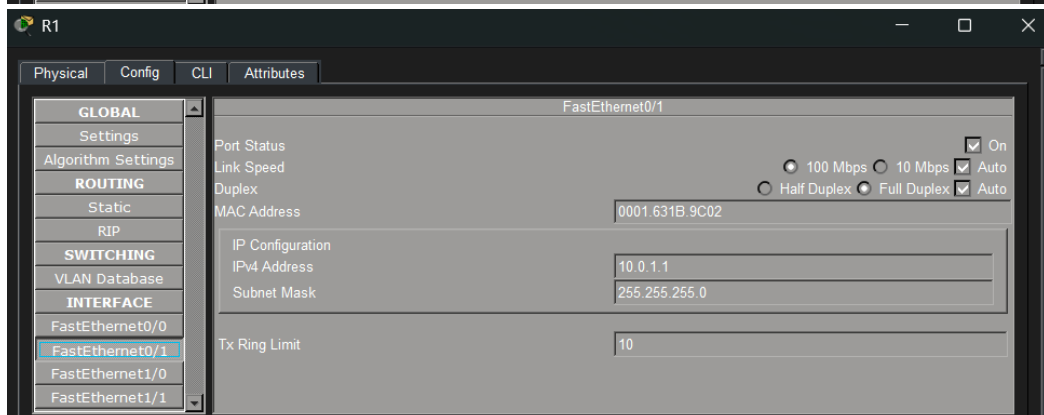
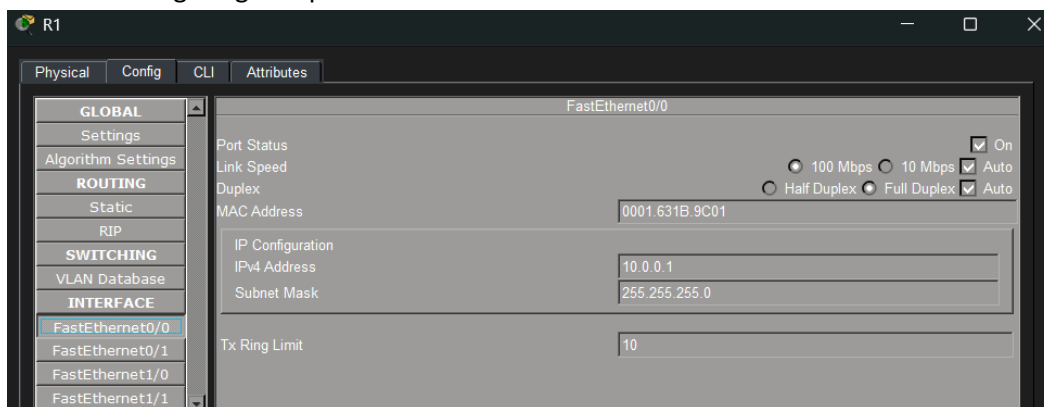
Topology pre tasks and no configurations.

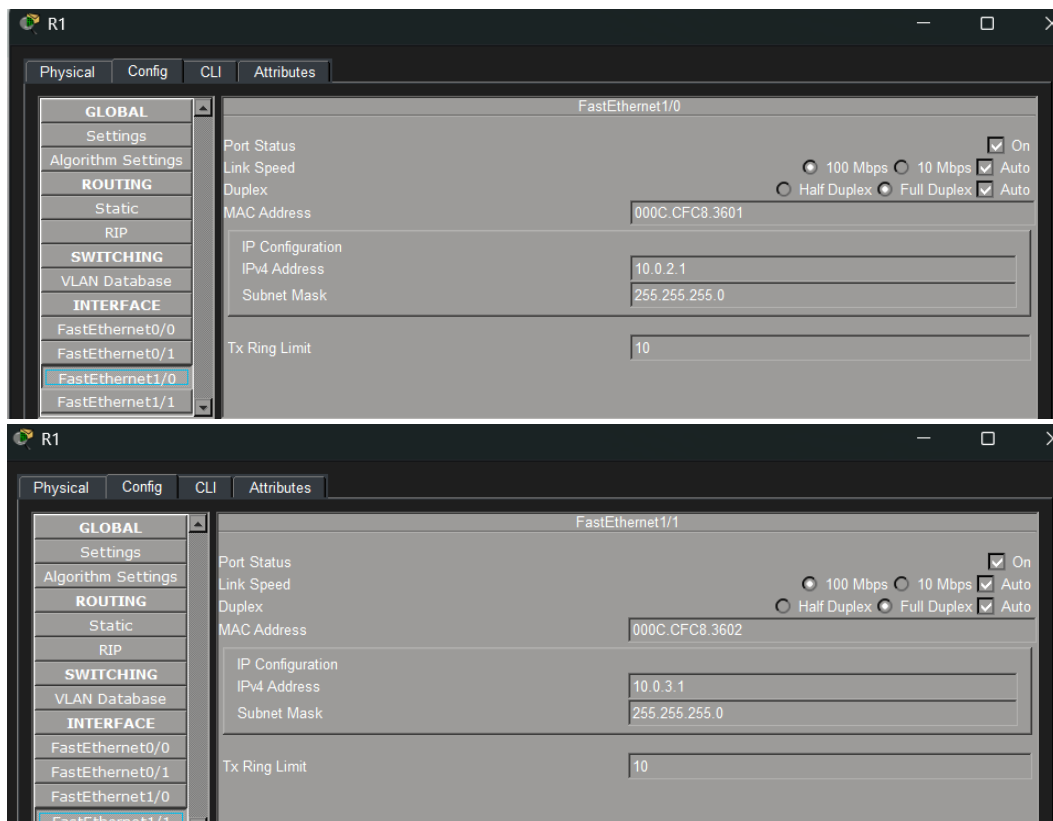
Task 52: Host names on all routers





Task 53: Configuring the ip routes





Task 54: I am unsure how to verify, especially when most aren't configured yet as per the order of tasks.

Task 55:

Tracert and ping on PC1 to PC2

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time=1ms TTL=127
Reply from 10.0.2.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  1  0 ms    1 ms     0 ms    10.0.1.1
  2  0 ms    1 ms     0 ms    10.0.2.10

Trace complete.

C:\>
```

Task 56: Host unreachable when pining pc1 to pc3, as the routers between are not linked.

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Task 57: Configure ip addresses on R2, R3 and R4. (Not the internet one)

```
ip address 10.0.0.2 255.0.0.0
R2(config-if)#ip address 10.0.0.2 255.255.255.0
R2(config-if)#
R2(config-if)#exit
R2(config)#interface FastEthernet0/1
R2(config-if)#no shutdown
R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
ip address 10.1.0.2 255.255.255.0
R2(config-if)#ip address 10.1.0.2 255.255.255.0
R2(config-if)#
R2(config-if)#
R2(config-if)#exit
R2(config)#interface FastEthernet0/0
R2(config-if)#
```

```

R3(config)#interface FastEthernet0/0
R3(config-if)#no shutdown
R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
ip address 10.1.0.1 255.0.0.0
R3(config-if)#ip address 10.1.0.1 255.255.255.0
R3(config-if)#
R3(config-if)#exit
R3(config)#interface FastEthernet0/1
R3(config-if)#no shutdown
R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

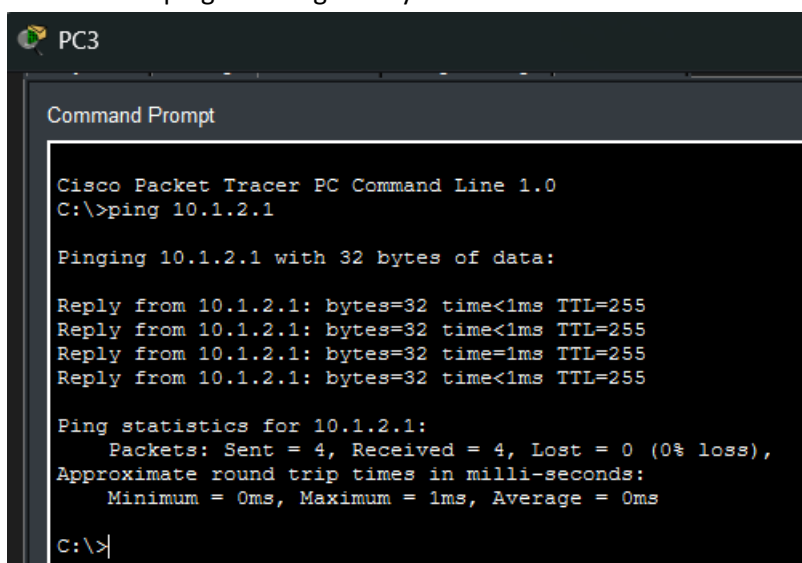
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 10.1.1.2 255.255.255.0
R3(config-if)#ip address 10.1.1.2 255.255.255.0
R3(config-if)#
Enter configuration commands, one per line. End with CNTRL/Z.
R4(config)#interface FastEthernet0/0
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.1.1.1 255.0.0.0
R4(config-if)#ip address 10.1.1.1 255.255.255.0
R4(config-if)#
R4(config-if)#exit
R4(config)#interface FastEthernet0/1
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 10.1.2.1 255.255.255.0
R4(config-if)#ip address 10.1.2.1 255.255.255.0
R4(config-if)#
R4(config-if)#exit
R4(config)#interface FastEthernet1/0
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
ip address 10.1.3.1 255.255.255.0
R4(config-if)#ip address 10.1.3.1 255.255.255.0
R4(config-if)#
R4(config-if)#exit
R4(config)#interface FastEthernet1/1
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up

```

Task 58: PC3 ping default gateway



The screenshot shows a PC3 window with a Command Prompt. The prompt is titled "PC3" and "Command Prompt". The text inside the prompt is as follows:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

59: Static routes on routers 1-4

```
R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#
Enter configuration commands, one per line. End with CNF
R2(config)#ip route 10.0.1.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.1
R2(config)#
Enter configuration commands, one per line. End with CNF
R3(config)#ip route 10.0.0.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.1.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.2.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.1.2.0 255.255.255.0 10.1.1.1
R4(config)#ip route 10.0.0.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.1.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.2.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.1.0.0 255.255.255.0 10.1.1.2
R4(config)#
```

Task 60: Connectivity between the 3 pcs.

```
C:\>tracert 10.1.2.10

Tracing route to 10.1.2.10 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.0.1.1
  2  0 ms      0 ms      0 ms      10.0.0.2
  3  0 ms      0 ms      0 ms      10.1.0.1
  4  0 ms      0 ms      0 ms      10.1.1.1
  5  10 ms     0 ms      0 ms      10.1.2.10

Trace complete.

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.0.1.1
  2  0 ms      0 ms      0 ms      10.0.2.10

Trace complete.

C:\>
```

Task 61: Remove static routes on R1, PC1 can no longer ping pc3

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.

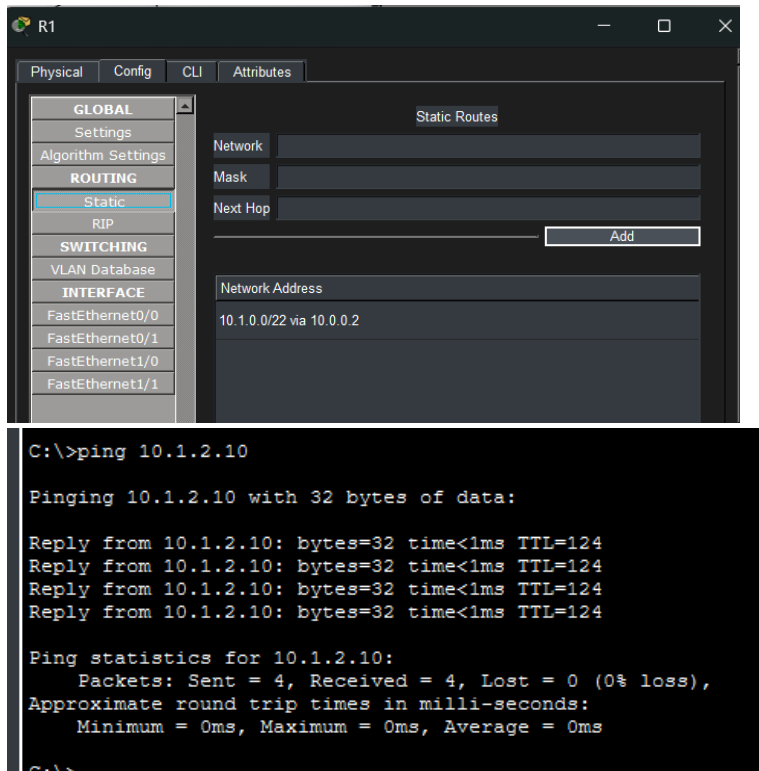
Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Task 62:

```
R1(config)#ip route 10.1.0.0 255.255.252.0 10.0.0.2
```

Task 63: Only one summary route now, PC1 can ping PC3 again.



The screenshot shows the R1 configuration window with the 'Static Routes' tab selected. The 'Network' field is set to '10.1.0.0/22 via 10.0.0.2'. Below this, the 'Network Address' field is also set to '10.1.0.0/22 via 10.0.0.2'. The 'Add' button is visible. Below the configuration window, a terminal window shows the command 'C:\>ping 10.1.2.10' and its output:

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124

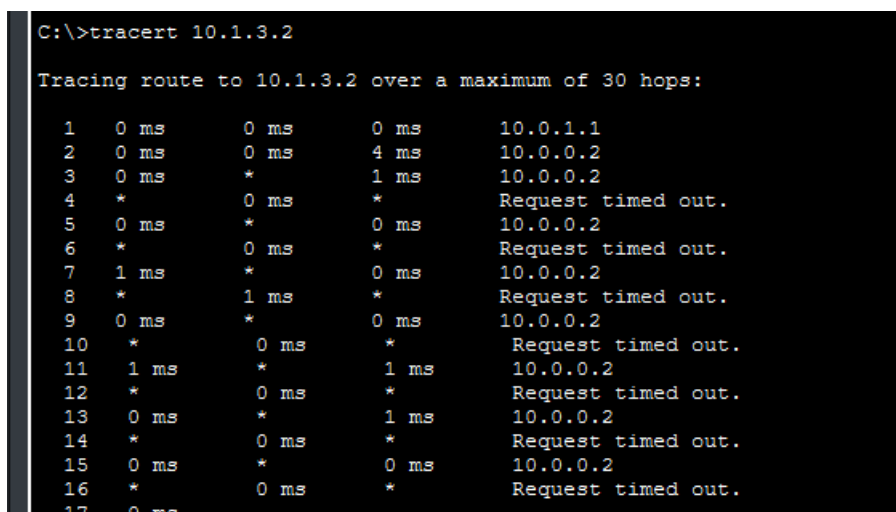
Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task 64

Device Name: R5			
Custom Device Model: 2811 IOS15			
Hostname: R5			
Port	Link	VLAN	IP Address
FastEthernet0/0	Up	--	10.1.3.2/24
FastEthernet0/1	Up	--	10.0.3.2/24

Task 65: PC1 tries to go through R1, to R2 and around all the way to R5, but fails.



The screenshot shows a terminal window with the command 'C:\>tracert 10.1.3.2' and its output:

```
C:\>tracert 10.1.3.2

Tracing route to 10.1.3.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.0.1.1
  2  0 ms    0 ms    4 ms    10.0.0.2
  3  0 ms    *        1 ms    10.0.0.2
  4  *        0 ms    *        Request timed out.
  5  0 ms    *        0 ms    10.0.0.2
  6  *        0 ms    *        Request timed out.
  7  1 ms    *        0 ms    10.0.0.2
  8  *        1 ms    *        Request timed out.
  9  0 ms    *        0 ms    10.0.0.2
 10  *        0 ms    *        Request timed out.
 11  1 ms    *        1 ms    10.0.0.2
 12  *        0 ms    *        Request timed out.
 13  0 ms    *        1 ms    10.0.0.2
 14  *        0 ms    *        Request timed out.
 15  0 ms    *        0 ms    10.0.0.2
 16  *        0 ms    *        Request timed out.
 17  0 ms
```

Task 66:

```
R1(config)#ip route 10.1.3.0 255.255.255.0 10.0.3.2
R1(config)#
```

Task 67: PC1 to R5's FE0/0 still does not connect.

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Enter configuration commands, one per line. End with Ctrl-Z.

```
R5(config)#ip route 10.0.1.0 255.255.255.0 10.0.3.1
R5(config)#
```

After the return path was made, connection was successful

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Reply from 10.1.3.2: bytes=32 time<1ms TTL=254
Reply from 10.1.3.2: bytes=32 time<1ms TTL=254
Reply from 10.1.3.2: bytes=32 time<1ms TTL=254
Reply from 10.1.3.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 10.1.3.2

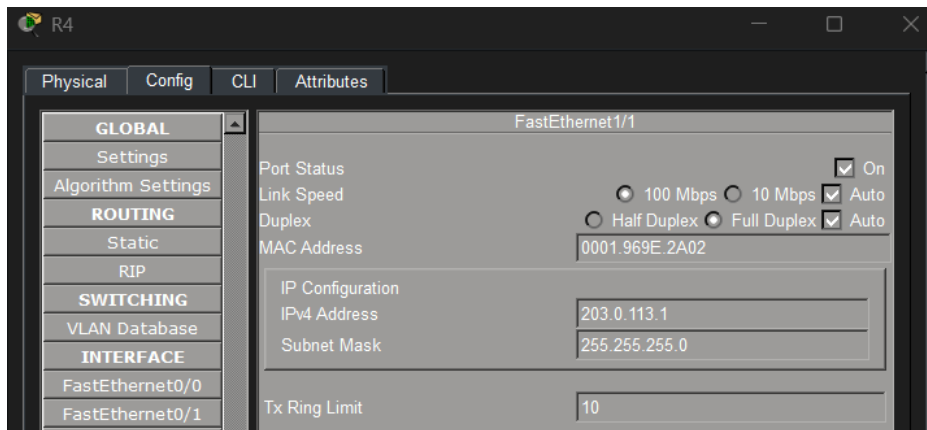
Tracing route to 10.1.3.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.1.1
  1  0 ms    0 ms    0 ms    10.1.3.2

Trace complete.

C:\>
```

Task 68:



Task 69:

```
R1(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.1
R2(config)#ip route 0.0.0.0 0.0.0.0 10.1.0.1
R3(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.1
R5(config)#ip route 0.0.0.0 0.0.0.0 10.1.3.1
```

Task 70:

```
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
```

Task 71: Every network is connected directly and routed, and I can see the routes via show ip route command. Where all of the connections are added to the routing table.

Task 72: The routes are the path the connection takes, and with the PC's not being directly connected to the same router, they need a table entry that they either learn or we give them to do the routing correctly.

Task 73: Summary routes basically route the whole block instead of single ports, which works the same way but works just as well without extra configuration if you add any more subnets.